Mausam Kar

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Education

VIT Bhopal University **September 2024 – May 2028**

Bachelor of Technology in Computer Science and Engineering, CGPA: 9.13/10.0 *Bhopal, India*

Experience

Data Science Intern **Present**

Prodigy Infotech *Remote*

- Performed end-to-end data analysis and preprocessing including data cleaning, feature engineering, and exploratory data analysis to extract actionable insights from large datasets
- Implemented machine learning models using Python, NumPy, Pandas, and Scikit-learn for supervised and unsupervised learning problems including classification and clustering tasks
- Developed and evaluated predictive models using performance metrics such as accuracy, precision, and recall, and created data visualizations using Matplotlib and Seaborn to support data-driven business decisions

Technical Skills

Programming Languages: C++, C, Java, Python, TypeScript, JavaScript
Web Technologies: React, Angular, Node.js, Express.js, HTML, CSS, Spring Boot
Mobile Development: React Native, Expo, NativeWind, Expo Router
Databases: MySQL, MongoDB, PostgreSQL, Prisma, Drizzle ORM, Appwrite
Tools and Technologies: Docker, Git, GitHub, RESTful APIs, Apache Kafka, Upstash Redis, WebSockets, Jest, Postman, Linux, AWS, Google Cloud Platform, Vercel, Render
Machine Learning: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Google Gemini, Pinecone, RAG
Other Skills: Data Structures and Algorithms, Object-Oriented Programming, Agile Methodology, CI/CD

Projects

Shikshak - Enterprise Learning Management System | *Next.js, Node.js, Kafka, MongoDB, Redis, Docker*

- Architected microservices-based LMS with event-driven design using Apache Kafka, enabling horizontal scaling and fault tolerance across 5 independent services (API Gateway, Auth, Courses, Payment, RAG)
- Implemented service decomposition strategy isolating compute-intensive operations (video processing, AI inference) into dedicated microservices for independent scaling based on workload
- Built RAG-powered study assistant with Redis vector database and LLM integration for semantic search and context-aware response generation across course materials
- Developed real-time AI proctoring using MediaPipe and TensorFlow.js for browser-based monitoring, and integrated Better-Auth for secure role-based access control

CampusSync - AI-Driven Educational Management Platform | *React, Node.js, MongoDB, AWS*

- Designed and developed an AI-driven serverless educational ERP system enabling real-time data synchronization, autonomous workflows, and personalized academic support for educational institutions
- Built a context-aware AI assistant for predictive student intervention, automated grading, and intelligent content generation using machine learning insights and natural language processing
- Architected a scalable event-driven system using serverless architecture supporting over 10,000 concurrent users with high availability and zero-operations deployment
- Published project report in International Journal and College Technical Journal

Real Scout - Real Estate Mobile Application | *React Native, Expo, TypeScript, Appwrite, NativeWind*

- Built cross-platform real estate mobile app using React Native and Expo with Google OAuth authentication and Appwrite backend for data management, authentication, and file storage
- Implemented dynamic property listings with advanced search and filtering by type, price range, and location with real-time data synchronization using Appwrite SDK
- Designed scalable architecture with clean separation of concerns, custom hooks for state management, and responsive UI using NativeWind (Tailwind CSS for React Native)

Achievements

Academic and Technical Achievements

- Published CampusSync project report in International Journal and College Technical Journal, 2024
- Secured 1st position in Interclub Hackathon by building a RAG-based AI system for context-aware information retrieval, 2024
- Implemented modern web animations using GSAP and Lenis libraries to enhance user interface and user experience with smooth scrolling and interactive transitions

Certifications

Applied Machine Learning with Python - University of Michigan (Coursera) *2024*

Leadership and Activities

- Linpack Club Core Member: Led 2 hackathons and organized 3 large-scale speaker sessions and technical workshops including fake currency detection using MATLAB, increasing member participation by 50 percent and new memberships by 30 percent